

Model Quality Self-Review Checklist

LUA BIM LABS Starter - Track 4 Quick Reference | Track completion card - Day 28

4-Area Quality Framework

Model quality is verified across four areas. Complete all 20 checks before submitting any MEP model.

Area 1: Geometry

- [v] All elements are placed at correct elevations using numeric offsets - not visually estimated
- [v] No elements floating in space, buried in a slab, or placed off-grid
- [v] All ducts and pipes have valid, confirmed sizes (not default or approximate)
- [v] Equipment is placed at correct position relative to structural grids
- [v] No duplicate elements at the same location (check with Manage > Review Warnings)

Area 2: Data / Parameters

- [v] All elements have a unique Mark value - no blanks, no duplicates
- [v] All elements are assigned to a named system (no "Default" or unnamed systems)
- [v] Required parameters are populated: Size, Level, System Name, Material/Specification
- [v] Equipment tags match the design drawing or equipment schedule
- [v] File name follows the project BEP naming convention

Area 3: System Connections

- [v] All duct and pipe ends are connected - no open connectors
- [v] The Systems Browser shows no unassigned or "unnamed" systems
- [v] Equipment (AHU, pump, panel) is connected to its associated ductwork or pipework
- [v] Gravity drainage is modeled with required slope (not flat) - typically 1:50 for drainage
- [v] System Inspector shows zero errors for each active system

Area 4: Coordination Readiness

- [v] All linked models are loaded and up to date (not broken links)
- [v] Shared coordinates confirmed - model aligns with architectural base
- [v] Model purged of unused families and types (File > Purge Unused)
- [v] Revit warnings reviewed - critical warnings resolved or documented
- [v] File synchronized to the central model before handover

LOD Self-Check Table

LOD	Geometry Confirmed	Parameters Populated	Hangers / Supports Include	Clash-Checked
LOD 200	Approximate	Minimal	No	No
LOD 300	Accurate	Core fields filled	No	Optional
LOD 350	Accurate	Full set required	Yes	Yes - required
LOD 400	Fabrication detail	All fields including	Yes (detailed)	Yes
LOD 500	As-installed	All fields and verification	As installed	N/A - record

Quick Reference: Common Errors to Check

Error	Where to Check
Open connectors	Systems Browser - unhosted elements

Wrong system type	Properties panel -> System Classification
Blank Mark	Schedule view, sort by Mark field
Warnings count	Manage tab -> Review Warnings
File size (large)	File > Purge Unused; check RVT file size